

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

All functional features included in the project solution, with purpose, status, and contribution.

S.NO	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Home Page	Introduces the project with a hero banner and background on flood forecasting	templates/home.html	Done	1
2	Prediction Form	Collects all 10 weather/rainfall readings required by the model	templates/predict.html	Done	1
3	Prediction Route	Validates form input, builds the feature DataFrame, and returns a result	app.py (POST /predict)	Done	1
4	Model Inference	Scales input and classifies flood risk using the trained model	floods.save, transform.save	Done	1
5	Result Page + Risk	Displays an animated SVG	chance.html,	Done	1

	Gauge	gauge reflecting prediction confidence/risk level	no_chance.html, main.js		
6	Light/Dark Theme Toggle	Lets users switch themes; choice persists via localStorage	main.css, main.js	Done	0.5
7	Ambient Rain/Lightning Animation	Canvas-based rain and lightning effects for atmosphere	main.js (initRain, initLightning)	Done	0.5
8	Deployment Config	Procfile + gunicorn setup for hosting on Render	Procfile, requirements.txt	Done	1

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Home, Form, Route, Inference, Result Page)
Additional / Nice-to-Have Features	3 (Theme toggle, Ambient animation, Deployment config)
Features Tested & Verified	8 (manually via browser, curl, and Postman)

Feature Category Breakdown

S.NO	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home page, Prediction form, Theme toggle
2	Backend / Logic	2	Flask routes, form validation
3	Machine Learning	1	Scaling + prediction via Joblib model
4	Presentation / Result	1	Animated risk gauge result pages
5	Deployment	1	Procfile + Gunicorn on Render